Solutions to UGB Mock 1

Q.1

Problem Let $f : [0, 1] \to \mathbb{R}$ be continuous, and suppose that $f(0) = f(1)$. Show that there is a point $a \in \left[0, \frac{2025}{2026}\right]$ satisfying $f(a) = f\left(a + \frac{1}{2026}\right)$.

Solution: Consider the function $g(x) = f\left(x + \frac{1}{2026}\right) - f(x)$, which is well-defined and continuous on the interval $\left[0, \frac{2025}{2026}\right]$. We evaluate the sum of $g(x)$ at 2026 evenly spaced points:

$$\sum_{k=0}^{2025} g\left(\frac{k}{2026}\right) = \sum_{k=0}^{2025} \left(f\left(\frac{k+1}{2026}\right) - f\left(\frac{k}{2026}\right)\right)$$

This sum telescopes, leaving only the boundary terms:

$$\sum_{k=0}^{2025} g\left(\frac{k}{2026}\right) = f(1) - f(0)$$

Since we are given $f(0) = f(1)$, the sum equals 0. For a sum of real numbers to be strictly zero, either all terms are identically zero, or there exist at least one non-negative term and at least one non-positive term. Therefore, there must exist integers $i, j \in [0, 2025]$ such that $g\left(\frac{i}{2026}\right) \le 0$ and $g\left(\frac{j}{2026}\right) \ge 0$.

Since $g(x)$ is a continuous function, the Intermediate Value Theorem guarantees the existence of some a between $\frac{i}{2026}$ and $\frac{j}{2026}$ (and thus $a \in \left[0, \frac{2025}{2026}\right]$) such that $g(a) = 0$. This implies $f\left(a + \frac{1}{2026}\right) - f(a) = 0$, which completes the proof. ■

Q.2

Problem: Given real numbers $0 = x_1 < x_2 < \cdots < x_{2026} < x_{2027} = 1$ with $x_{i+1} - x_i \le \frac{1}{2}$ for $1 \le i \le 2026$. Show that $\frac{1}{4} < \sum_{i=1}^{1013} x_{2i}(x_{2i+1} - x_{2i-1}) < \frac{3}{4}$.

Solution: Let $S = \sum_{i=1}^{1013} x_{2i}(x_{2i+1} - x_{2i-1})$. Using the identity $2xy = (y + x)y - (y - x)y$, we can rewrite the terms by introducing squares. Specifically:

$$x_{2i}(x_{2i+1} - x_{2i-1}) = \frac{1}{2}\left(x_{2i+1}^2 - x_{2i-1}^2\right) - \frac{1}{2}\left(x_{2i+1} - x_{2i}\right)^2 + \frac{1}{2}\left(x_{2i} - x_{2i-1}\right)^2$$

Let $\Delta_k = x_{k+1} - x_k$. Summing the above expression from $i = 1$ to 1013 creates a telescoping sum for the squared terms:

$$S = \frac{1}{2}\sum_{i=1}^{1013}\left(x_{2i+1}^2 - x_{2i-1}^2\right) + \frac{1}{2}\sum_{i=1}^{1013}\left(\Delta_{2i-1}^2 - \Delta_{2i}^2\right)$$

The first summation evaluates to $\frac{1}{2}(x_{2027}^2 - x_1^2) = \frac{1}{2}(1^2 - 0^2) = \frac{1}{2}$. Thus,

$$S = \frac{1}{2} + \frac{1}{2}\sum_{i=1}^{1013} \Delta_{2i-1}^2 - \frac{1}{2}\sum_{i=1}^{1013} \Delta_{2i}^2$$

Since $\Delta_k > 0$ and $\sum_{k=1}^{2026} \Delta_k = 1$, we know that $\sum_{i=1}^{1013} \Delta_{2i-1} < 1$ and $\sum_{i=1}^{1013} \Delta_{2i} < 1$. We are given $\Delta_k \leq \frac{1}{2}$. Thus, for any sub-sum of squares:

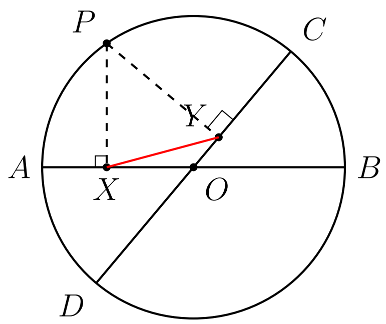
$$\sum \Delta_{2i-1}^2 < (\max \Delta_{2i-1}) \sum \Delta_{2i-1} \leq \frac{1}{2} \times 1 = \frac{1}{2}$$

$$\sum \Delta_{2i}^2 < (\max \Delta_{2i}) \sum \Delta_{2i} \leq \frac{1}{2} \times 1 = \frac{1}{2}$$

Therefore, the error term $E = \frac{1}{2}\sum \Delta_{2i-1}^2 - \frac{1}{2}\sum \Delta_{2i}^2$ is strictly bounded between $-\frac{1}{4}$ and $\frac{1}{4}$. Substituting this back yields $\frac{1}{2} - \frac{1}{4} < S < \frac{1}{2} + \frac{1}{4}$, proving $\frac{1}{4} < S < \frac{3}{4}$. ■

Q.3

Problem: Let AB and CD be two given diameters of a given circle with centre O. Let P be a point chosen arbitrarily on the circle. Drop perpendiculars from P to the two diameters, meeting them at X and Y. Prove that the length of XY does not change no matter where P is placed on the circle.



Solution: Since $PX \perp AB$ and $PY \perp CD$, the angles $\angle OXP$ and $\angle OYP$ are both exactly $90°$. Because $\angle OXP + \angle OYP = 180°$, the points O, X, P, and Y are concyclic (they lie on the same circle). The diameter of this new circle is OP because the angle subtended by OP at X (and Y) is $90°$.

Since P lies on the original circle with center O, the length OP is simply the radius of the original circle, which we denote as R. In this new circle, XY is a chord that subtends the angle $\angle XOY$ at the circumference. Note that $\angle XOY$ is the fixed angle between the two given diameters AB and CD. Let this fixed angle be θ.

By the Law of Sines on the circle containing O, X, P, Y:

$$\frac{XY}{\sin\theta} = \text{Diameter} = OP = R$$

Thus, $XY = R\sin\theta$. Since R and θ are constants defined entirely by the fixed geometry of the initial circle and diameters, the length of XY is constant regardless of the position of P. ■

Q.4

Problem: Let $a, b \in \mathbb{R}$ and $f(x) = a\cos x + b\sin 3x$. It is given that $f(x) > 1$ has no solutions in real. Prove that $|b| \leq 1$.

Solution: Since the inequality $f(x) > 1$ has no real solutions, it must be true that $f(x) \leq 1$ for all $x \in \mathbb{R}$. We can selectively evaluate the function at specific points to isolate b.
Let $x = \frac{\pi}{2}$. We obtain:

$$f\left(\frac{\pi}{2}\right) = a\cos\left(\frac{\pi}{2}\right) + b\sin\left(\frac{3\pi}{2}\right) = a(0) + b(-1) = -b$$

Since $f(x) \leq 1$ for all x, we have $-b \leq 1$, which implies $b \geq -1$.
Next, let $x = \frac{3\pi}{2}$. We obtain:

$$f\left(\frac{3\pi}{2}\right) = a\cos\left(\frac{3\pi}{2}\right) + b\sin\left(\frac{9\pi}{2}\right) = a(0) + b(1) = b$$

Applying the global condition again, $b \leq 1$.
Combining both inequalities, we get $-1 \leq b \leq 1$, which is equivalent to $|b| \leq 1$. ■

Q. 5

Problem: Let x_1, x_2 be the two roots of the equation $x^2 + px - 1 = 0$ where p is an odd integer. Let $y_n = x_1^n + x_2^n$ for $n \geq 0$. Show that y_n, y_{n+1} are coprime for all integers $n \geq 0$.

Solution: By Vieta's formulas, $x_1 + x_2 = -p$ and $x_1x_2 = -1$. We can establish a linear recurrence relation for y_n:

$$y_{n+1} = x_1^{n+1} + x_2^{n+1} = (x_1 + x_2)(x_1^n + x_2^n) - x_1x_2(x_1^{n-1} + x_2^{n-1})$$

Substituting our values from Vieta's formulas, we get:

$$y_{n+1} = -py_n - (-1)y_{n-1} \implies y_{n-1} = y_{n+1} + py_n$$

Let $d = \gcd(y_n, y_{n+1})$. From the recurrence relation, any divisor of both y_n and y_{n+1}

must also divide y_{n-1}. By Euclidean algorithm, $\gcd(y_{n+1}, y_n) = \gcd(y_n, y_{n-1})$.

Applying this recursively down to the base cases gives $\gcd(y_n, y_{n+1}) = \gcd(y_1, y_0)$. Evaluating the base cases:

$$y_0 = x_1^0 + x_2^0 = 1 + 1 = 2$$

$$y_1 = x_1^1 + x_2^1 = -p$$

Thus, $\gcd(y_n, y_{n+1}) = \gcd(-p, 2) = \gcd(p, 2)$. Since it is given that p is an odd integer, $\gcd(p, 2) = 1$. Therefore, y_n and y_{n+1} are always coprime. ∎

Q. 6

Problem: Let f be continuous and monotonically increasing, with $f(0) = 0$, and $f(1) = 1$. Prove that $\sum_{k=1}^{9} f\left(\frac{k}{10}\right) + \sum_{k=1}^{9} f^{-1}\left(\frac{k}{10}\right) \leq \frac{99}{10}$.

Solution: Consider the 2D Cartesian plane. For each $k \in \{1, \ldots, 9\}$, the term $\frac{1}{10} f\left(\frac{k}{10}\right)$ represents the area of the rectangle $A_k = \left[\frac{k}{10}, \frac{k+1}{10}\right] \times \left[0, f\left(\frac{k}{10}\right)\right]$. Because f is strictly increasing, A_k lies entirely on or below the curve $y = f(x)$. Furthermore, the union $A = \bigcup_{k=1}^{9} A_k$ is contained entirely within the bounding box $[0.1, 1] \times [0, 1]$.

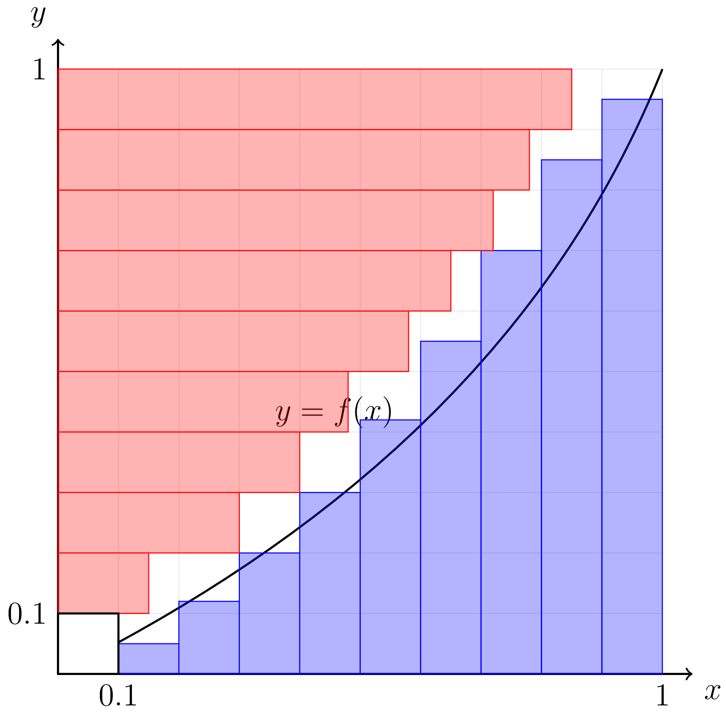
Similarly, the term $\frac{1}{10} f^{-1}\left(\frac{j}{10}\right)$ represents the area of the rectangle $B_j = \left[0, f^{-1}\left(\frac{j}{10}\right)\right] \times \left[\frac{j}{10}, \frac{j+1}{10}\right]$. This region lies on or strictly to the left of the curve $x = f^{-1}(y)$, which translates to being on or above the curve $y = f(x)$. The union $B = \bigcup_{j=1}^{9} B_j$ is contained entirely within $[0, 1] \times [0.1, 1]$.

Since A is below the curve and B is above it, their interiors are disjoint. Thus, the sum of their areas equals the area of their union. The union $A \cup B$ is strictly bounded by the shape formed by the union of their bounding boxes: $([0.1, 1] \times [0, 1]) \cup ([0, 1] \times [0.1, 1])$. This combined region represents the entire unit square $[0, 1] \times [0, 1]$ minus the single bottom-left 0.1×0.1 square $[0, 0.1] \times [0, 0.1]$.

The area of this geometric envelope is $1 - (0.1 \times 0.1) = 0.99$. Consequently, the sum of the areas is ≤ 0.99:

$$\frac{1}{10} \sum_{k=1}^{9} f\left(\frac{k}{10}\right) + \frac{1}{10} \sum_{k=1}^{9} f^{-1}\left(\frac{k}{10}\right) \leq 0.99$$

Multiplying by 10 yields the requested bound $\leq 9.9 = \frac{99}{10}$. Equality holds when $f(x) = x$ because the rectangles expand to fill the maximal region perfectly. ∎



Q.7

Problem: Suppose by $[\cdot]$, we denote the greatest integer function. Answer the following:

(a) For all $x, y \in \mathbb{R}$ show that $[2x] + [2y] \geq [x] + [y] + [x + y]$.

(b) Using the previous result or otherwise, prove that $\frac{\binom{2m}{m}\binom{2n}{n}}{\binom{m+n}{n}}$ is an integer for all $m, n \in \mathbb{N}$.

Proof of (a): Let $x = n_x + f_x$ and $y = n_y + f_y$, where $n_x, n_y \in \mathbb{Z}$ and $f_x, f_y \in [0, 1)$. Substituting these into the inequality simplifies it to verifying $[2f_x] + [2f_y] \geq [f_x + f_y]$ because the integer parts cancel uniformly on both sides. We analyze the four possible boundary cases for the fractional parts:

1. $f_x < 0.5, f_y < 0.5$: LHS $= 0 + 0 = 0$. RHS $= [f_x + f_y] = 0$. $(0 \geq 0)$

2. $f_x \geq 0.5, f_y < 0.5$: LHS $= 1 + 0 = 1$. RHS $= [f_x + f_y] \leq 1$. $(1 \geq 1)$

3. $f_x < 0.5, f_y \geq 0.5$: By symmetry, LHS $= 1$, RHS ≤ 1. $(1 \geq 1)$

4. $f_x \geq 0.5, f_y \geq 0.5$: LHS $= 1 + 1 = 2$. RHS $= [f_x + f_y] \leq 1$ since $f_x + f_y < 2$. $(2 \geq 1)$

The inequality holds in all cases. ■

Proof of (b): Let $N = \frac{\binom{2m}{m}\binom{2n}{n}}{\binom{m+n}{n}}$. Expanding the binomial coefficients into factorials yields:

$$N = \frac{\frac{(2m)!}{(m!)^2}\frac{(2n)!}{(n!)^2}}{\frac{(m+n)!}{m!n!}} = \frac{(2m)!(2n)!}{m!n!(m+n)!}$$

To show N is an integer, we must prove that the p-adic valuation $v_p(N) \geq 0$ for every prime p. Using Legendre's formula, the valuation of the factorials is:

$$v_p(N) = \sum_{k=1}^{\infty} \left(\left\lfloor \frac{2m}{p^k} \right\rfloor + \left\lfloor \frac{2n}{p^k} \right\rfloor - \left\lfloor \frac{m}{p^k} \right\rfloor - \left\lfloor \frac{n}{p^k} \right\rfloor - \left\lfloor \frac{m+n}{p^k} \right\rfloor \right)$$

Let $x = \frac{m}{p^k}$ and $y = \frac{n}{p^k}$. The inner term of the sum perfectly matches the inequality established in part (a): $[2x] + [2y] - [x] - [y] - [x+y] \geq 0$. Since each term in the infinite series is non-negative, $v_p(N) \geq 0$ for all primes p. Thus, N contains no fractions in its prime factorization and must be an integer. ■

Q.8

Problem: The length of each side of a convex quadrilateral $ABCD$ is strictly less than 24. Let P be any point inside $ABCD$. Prove that there is at least one vertex among A, B, C, D such that its distance from P is strictly less than 17.

Solution: Let the distances from P to the vertices A, B, C, D be PA, PB, PC, and PD. We proceed by contradiction. Assume that all four distances are at least 17 (i.e., $PA, PB, PC, PD \geq 17$).

Point P is inside the convex quadrilateral, so it subtends four angles that complete a full circle: $\angle APB, \angle BPC, \angle CPD$, and $\angle DPA$. Their sum is exactly $360°$. By the Pigeonhole Principle, at least one of these four angles must be $\geq 90°$. Without loss of generality, assume $\angle APB \geq 90°$.

Applying the Law of Cosines to $\triangle APB$:

$$AB^2 = PA^2 + PB^2 - 2(PA)(PB)\cos(\angle APB)$$

Since $\angle APB \geq 90°$, its cosine is non-positive ($\cos(\angle APB) \leq 0$). Thus:

$$AB^2 \geq PA^2 + PB^2$$

Substituting our assumption that $PA \geq 17$ and $PB \geq 17$:

$$AB^2 \geq 17^2 + 17^2 = 289 + 289 = 578$$

However, we are explicitly given that every side length of the quadrilateral is strictly less than 24. This requires:

$$AB < 24 \implies AB^2 < 576$$

We have derived that $578 \leq AB^2 < 576$, which is an obvious contradiction. Thus, our initial assumption must be false, proving that at least one vertex must have a distance from P that is strictly less than 17. $\blacksquare$
